
FACE
Prep

Mensuration

FACE Prep

Question

A park in the form of a right-angled triangle has a base and height of 10 m and 15 m respectively. Find the area of the park?

A) 45

B) 75

C) 60

D) 150

FACE Prep

Question

Let the length of the side be $4\sqrt{3}$ cm.

A) What is the height of an equilateral triangle?($\sqrt{3}a/2$)

B) What is the area of an equilateral triangle?($\sqrt{3}/4\ a^2$)

C) What is the area of a regular hexagon?($3\sqrt{3}/2 * a^2$)

D) What is the perimeter of a regular hexagon(6a)

FACE Prep

Question

Mr. Frankenstein wants to build a house with 7 rooms each having equal area. The length of each room is 60 m, breadth is 50 m and height is 20 m. For every unit of area, he requires 120 bricks. Tiles with the dimensions 3 m x 2 m are fixed only on the walls.

A) What is the floor area in each room?

B) What is the volume of each room?

C) How many tiles are required to make the walls of a particular room?

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Question

Let the diameter be 14 cm and height be 6 cm.

A) What is the volume of the cylinder?

B) What is the volume of the cone?

C) What is the total surface area of the cylinder?

FACE Prep

Question

A triangle is made from a rope. The sides of triangle are 19 cm, 14 cm and 31 cm. What will be the area of the square made from the rope?

FACE Prep

Question

How many bricks each of dimension 13 cm x 12 cm x 19 cm are required to construct a wall of dimensions 390 cm x 400 cm x 410 cm, if the wall constitutes 24% of sand and cement?

A) 20,000

B) 14,800

C) 18,200

D) 16,400

FACE Prep

Question

Find the height of a cylinder whose radius is 7 cm and the total surface area is 968 cm^2.

A) 15 cm

B) 14 cm

C) 12 cm

D) 16 cm

FACE Prep

Question

Find the height of a cuboid whose volume is 275 cm^3 and base area is 25 cm^2.

A) 17

B) 9

C) 11

D) 13

FACE Prep

Question

A rectangular piece of paper 11 cm × 4 cm is folded without overlapping to make a cylinder of height 4 cm. Find the volume of the cylinder.

A) $27.5cm^3$

B) $38.5cm^3$

C) $25.3cm^3$

D) $11.5cm^3$

FACE Prep

Question

The area of a trapezium shaped field is 480 m^2, the distance between two parallel sides is 15 m and one of the parallel sides is 20 m. Find the other parallel side.

A) 40m

B) 42m

C) 39m

D) 44m

FACE Prep

Question

A rhombus having diagonals of length 10 cm and 16 cm, respectively. Find its area.

A) $80cm^2$

B) $77cm^2$

C) $84cm^2$

D) $82cm^2$

FACE Prep

Question

Suppose a quadrilateral having a diagonal of length 10 cm, which divides the quadrilateral into two triangles and the heights of triangles with diagonals as the base, are 4 cm and 6 cm. Find the area of the quadrilateral.

A) $40cm^2$

B) $45cm^2$

C) $50cm^2$

D) $55cm^2$

FACE Prep

Question

If each edge of a cube is increased by 10 %, what would be the percentage increase in volume?

A) 33.1%

B) 11.1%

C) 22.1%

D) 25.5%

FACE Prep

Question

From a cylindrical vessel having a radius of the base of 7 cm and a height 6cm, water is poured into small hemispherical bowls each of radius 3.5 cm. Find the minimum number of bowls that would be required to empty the cylindrical vessel.

A) 13

B) 11

C) 14

D) 16

FACE Prep

Question

A solid metallic sphere of radius 26 cm is melted and recast into a cone having diameter of the base as 26 cm. What is the height of the cone?

A) 200 cm

B) 392 cm

C) 416 cm

D) 304 cm

FACE Prep

Question

The length of the edge of a cube is 6.3 cm. What will be the volume of the largest sphere that can be carved out from the cube?

A) 156.328 cm^3

B) 130.977 cm^3

C) 160.321 cm^3

D) 172.324 cm^3

FACE Prep

Question

The radius and height of right circular cylindrical pipe are 21 cm and 7 cm. Find the cost of overlaying the curved part of the pipe with aluminium sheet costing Rs. 2 per cm2

A) 1848

B) 422

C) 844

D) 924

FACE Prep

Question

The ratio of the length and the breadth of a rectangle is 4 : 3 and the area of the rectangle is 6912 sq cm. Find the ratio of the breadth and the area of the rectangle?

A) 1:96

B) 1:48

C) 1:84

D) 1:64

FACE Prep

Question

The radius of a wheel is 22.4 cm. What is the distance covered by the wheel in making 500 revolutions?

A) 252m

B) 704m

C) 352m

D) 808m

FACE Prep

Question

A park in the form of a right-angled triangle has a base and height of 10 m and 15 m respectively. Find the area of the park?

A) 45

B) 75

C) 60

D) 150

FACE Prep

Question

If a circle and a square have the same area, then what will be the perimeter of the square, if the diameter of the circle is 8 cm.

A) $4\sqrt{\pi}$

B) 3π

C) 8π

D) $16\sqrt{\pi}$

FACE Prep

Question

Find the length of the longest pole that can be placed in a room 12 m long, 8m broad and 9 m high.

A) 16m

B) 17m

C) 18m

D) 9m

FACE Prep

Question

If the radius of a cylinder is doubled and the height remains same, the volume will be

A) Doubled

B) Halved

C) same

D) Quadrupled

FACE Prep

Question

A polygon has 27 diagonals. The number of sides of the polygon is

A) 9

B) 10

C) 11

D) 12

FACE Prep

Question

A wire in the form of a circle of radius 3.5 m is bent in the form of a rectangle, whose length and breadth are in the ratio of 6 : 5. What is the area of the rectangle?

A) 60 sq.cm

B) 30 sq.cm

C) 45 sq.cm

D) 15 sq.cm

FACE Prep

Question

A contractor undertakes the job of fencing a rectangular field of length 100 m and breadth 50 m. The cost of fencing is Rs. 2 per meter and the labor charges are Re. 1 per meter, both paid directly to the contractor. Find the total cost of fencing if 10 % of the amount paid to the contractor is paid as tax to the land authority.

A) 1000

B) 580

C) 360

D) 990

FACE Prep

Question

If the length of the side of the rhombus is 10 cm and the length of the shorter diagonal is 12 cm then what is the area of the rhombus?

A) 96 sq.cm

B) 108 sq.cm

C) 80 sq.cm

D) 144 sq.cm

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Question

Water flows into a tank 200 m × 150 m through a rectangular pipe 1.5 m × 1.25 m @ 20 kmph. In what time (in minutes) will the water rise by 2 meters?

A) 234 min

B) 1.2 hours

C) 9 hours

D) 96 min

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Question

A hemispherical bowl of internal radius 9 cm contains a liquid. This liquid is to be filled into cylindrical shaped small bottles of diameter 3 cm and height 4 cm. How many bottles will be needed to empty the bowl?

A) 23

B) 34

C) 54

D) 46

FACE Prep

Question

Two metallic right circular cones having their heights 4.1 cm and 4.3 cm and the radii of their bases 2.1 cm each, have been melted together and recast into a sphere. Find the diameter of the sphere.

A) 4 cm

B) 4.1 cm

C) 4.2 cm

D) 4.3 cm

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Question

A cube of edge 15 cm is immersed completely in a rectangular vessel containing water. If the dimensions of the base of the vessel are 20 cm × 15 cm, find the rise in water level.

A) 11 cm

B) 11.11 cm

C) 11.22 cm

D) 11.25 cm

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Question

Which of the following cannot be the ratio of the angles of a right -angled triangle :

A) 1 : 2 :3

B) 1 : 1 : 2

C) 5 : 6 : 7

D) 1 : 5 : 6

FACE Prep

Question

The length of a rectangle is three fifth of the side of a square. The radius of a circle is equal to side of the square. The circumference of the circle is 132 cm. What is the area of the rectangle if the breadth of the rectangle is 8 cm?

A) 112.4 sq.cm

B) 104.2 sq.cm

C) 100.8 sq.cm

D) 109.3 sq.cm

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Question

The radius of a circular field is equal to the side of a square field. If the difference between the perimeter of the circular field and that of the square field is 32m, what is the perimeter of the square field? (in meter)

A) 84

B) 56

C) 95

D) 28

FACE Prep

Question

The length, breadth and height of a room are 7m,4m and 9m respectively. Find the cost of whitewashing the walls of the room and the ceiling at the rate of rs7.50 per m^2

A) rs. 1965

B) rs. 1695

C) rs. 1968

D) rs. 1698

FACE Prep

Question

A rectangular piece of paper is 52cm long and 22cm wide. A cylinder is formed by rolling the paper along its breadth. Find the volume of the cylinder.

A) 1870 cm^3

B) 2002 cm^3

C) 1030 cm^3

D) 1290 cm^3

FACE Prep

Question

Find the total surface area of a cylinder with diameter of base 28cm and height 70cm

A) $7300cm^2$

B) $7932\ cm^2$

C) $8000\ cm^2$

D) $7392\ cm^2$

FACE Prep

Question

The radius of a roller is 14 cm and its length 20cm . It takes 235 complete revolutions to move once over to evel a playground. Find the area of the playground.

A) $4136m^2$

B) $41360cm^2$

C) $413600cm^2$

D) $4136000\ cm^2$

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Question

The diameter of the driving wheel of a bus is 280 cm. How many revolutions per minute must the Wheel make in order to keep a speed of 66 kmph?

A) 150

B) 180

C) 145

D) 125

FACE Prep

Question

The area of a triangle is 216cm2 and its sides are in the ratio 3:4:5. The perimeter of the triangle is

A) 6 cm

B) 12 cm

C) 36 cm

D) 72 cm

FACE Prep

Question

In measuring the sides of a rectangle, one side is taken 5% in excess and other 4% in deficit. Find the error per cent in area calculated from the measurement.

A) -0.5%

B) 0.8%

C) 1%

D) -1%

FACE Prep

Question

The radius of a sphere is equal to the side of a cube having volume 729 cm3. What is the total surface area of the sphere? (Take $\pi = 3$)

A) 680 cm^2

B) 1080 cm^2

C) 606 cm^2

D) 972 cm^2

FACE Prep

Question

A circular park has been fenced at the rate of Rs. 6 per m. If the total cost of fencing is Rs. 3168, then find the diameter of the circular park.

A) 156 m

B) 172 m

C) 160 m

D) 168 m

FACE Prep

Question

The radius of a sphere is equal to the side of a cube having volume 729 cm3. What is the total surface area of the sphere? (Take $\pi = 3$)

A) 680 cm^2

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C) 606 cm^2

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FACE Prep

Question

The sum of areas of two rectangles (R1 and R2) of same length is 432 cm2, and the ratio of breadth of rectangle R1 and breadth of rectangle R2 is 5:4, respectively. If the area of square having length of side equal to the length of rectangle is 256 cm2, then find the area of square having length of side equal to the breadth of rectangle R2.

A) 196 cm^2

B) 100 cm^2

C) 144 cm^2

D) 256 cm^2

FACE Prep

Question

If the length of the rectangle is 8 cm more than the breadth of the rectangle and the perimeter of the rectangle is 64 cm. If the area of a square is equal to 15 times the area of the rectangle, then what is the perimeter of the square?.

A) 490 cm

B) 240 cm

C) 250 cm

D) 640 cm

FACE Prep

Question

The perimeter of the square is 112 m and the length of rectangle is equal to the side of the square. The area of the rectangle is 392 Sq m. Find the area of the circle whose radius is equal to the breadth of the rectangle?

A) 1386 sq.cm

B) 1078 sq. cm

C) 784 sq.cm

D) 616 sq. cm

FACE Prep

Question

If the circumference of the semicircle is 36 cm and the breadth of the rectangle is one-eleventh of the area of the semicircle. If the area of the rectangle is 168 cm^2, then what is the perimeter of the rectangle?

A) 62 cm

B) 58. cm

C) 44 cm

D) 56 cm

FACE Prep

Question

The diagonal of a square is 35 √2 cm, whose area is 161 Sq cm less than the area of a circle. Find the circumference of the circle?

A) 144 cm

B) 128 cm

C) 132 cm

D) 156 cm

FACE Prep

Question

If the volume and curved surface area of a cylinder are $616m^3$ and $352m^2$ respectively. what is the total surface area of the cylinder (in m^2)

A) 429

B) 419

C) 435

D) 421

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THANK YOU



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